II B.E. (Computer Engineering)
Class Test I August 2023
CER3G1: Computer Organization and Architecture

Time : 70 Min. **Max. Marks 20**

Note: Attempt as much as you can. Maximum marks awarded will not exceed 20.

Q1 Consider computer A having CPU processor as Pentium with 2.4GHz clock speed and computer B having CPU processor as Pentium pro with 1.2 Ghz clock rate. Assume that all other configuration of both computers is same. Which computer will be faster and why? **(6)**

Q. 2 A program is run on a 400 MHz processor. The executed program consist of the following instruction mix and clock cycle **(6)**

Instruction Type	Instruction Count	**Cycles per Instructio n**
Data Transfer	32,000	1
Integer Arithmetic	45,000	2
Floating Point Arithmetic	8,000	2
Branch	5,000	2

(i) Determine the effective CPI, MIPS rate, and execution time.
(ii) If the required MIPS is 250, would you recommend this processor to use as CPU in the computer. Give proper justification for your answer.

Q. 3 What is cache memory? Which cache memory among L1, L2 and L3 is faster and why? **(4)**

Q. 4 *What are the structural components of a microprocessor and a microcontroller? Why a processor of 32bit word length is faster than a processor of 16 bit word length.*

Q. 5 *Give correct option in the following : (+1mark for correct option, - 0.5 mark for incorrect option)* **(4)**

(i) Activities included in computer architecture
(a) interconnecting CPU, Memory and I/O devices (b) Deciding Bus width
(c) Selecting a processor as CPU (d) None of these

(ii) Intel's first microprocessor supporting parallel computing
(a) Pentium (b) Core (c) 80486 (d) None of these

(iii) Process to compare the performance of computers is called
(a) benchmarking (b) block chaining (c) Compiling (d) None of these

(iv) Example of Volatile memory
(a) RAM (b) ROM (c) Magnetic memory (d) virtual memory

Q. 6 *Name the register (MAR,MBR, IBR, IR, PC etc.) to store the following* **(4)**
(i)Memory address (ii) OPCODE (iii) Address of next instruction to be executed
(iv) Results of ALU operation
(+1mark for correct option, - 0.5 mark for incorrect option)